Calabi–Yau Quantum Groups

Chiral Algebras from Calabi–Yau Categories
via E_1/E_2 Factorization

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Contents

Part I

Calabi–Yau Categories and Cyclic Structures

Chapter 1

Introduction

1.1 The question

What is a Calabi–Yau quantum chiral algebra?

A Calabi–Yau category C of dimension d carries a canonical cyclic A_∞-structure: a non-degenerate trace

$$\mathrm{Tr}\colon \mathrm{HH}_\bullet(C) \longrightarrow k[-d]$$

on its Hochschild homology, encoding the CY condition as a d-dimensional Frobenius structure at the chain level. The cyclic bar complex $\mathrm{CC}_\bullet(C)$ is the primary invariant; it carries an $\mathbb{S}^d$-action (from the d-sphere framing of the trace), and its $\mathbb{S}^d$-equivariant structure governs higher-genus amplitudes.

A chiral algebra A on a curve X carries a bar complex $B(A)$, which is a factorization coalgebra on $\mathrm{Ran}(X)$. The bar differential encodes holomorphic OPE residues; the coproduct encodes topological interval-cutting. Together they assemble into a Swiss-cheese algebra structure (Volume II). At genus $g \geq 1$, the bar complex acquires curvature $\kappa(A) \cdot \omega_g$ from the Hodge bundle, and the full modular structure is controlled by the universal Maurer–Cartan element Θ_A (Volume I).

The programme of this work is to construct a precise bridge:

$$\left\{ \begin{array}{c} \text{CY categories with} \\ \text{appropriate } E_n\text{-structure} \end{array} \right\} \xrightarrow{\quad \Phi \quad} \left\{ \begin{array}{c} \text{quantum chiral algebras} \\ \text{with modular trace} \end{array} \right\}$$

The functor Φ should:

(i) take a CY category C (Fukaya, derived, matrix factorization, or more general) as input;

(ii) extract the E_2-monoidal structure on C (braided tensor product, quantum group action);

(iii) produce a chiral algebra A_C whose bar complex $B(A_C)$ encodes the CY cyclic homology;

(iv) realize the CY trace as the modular characteristic $\kappa(A_C)$.

1.2 The E_1/E_2 chiral hierarchy

The key structural ingredient is the E_n-chiral hierarchy:

- E_1-chiral algebras (Volume II, Part VII): associative factorization on $\mathbb{C} \times \mathbb{R}$, encoding the ordered/topological sector. The representation category of an E_1-chiral algebra is monoidal.
- E_2-chiral algebras (this work): braided factorization on $\mathbb{C} \times \mathbb{C}$, encoding the full holomorphic-holomorphic sector. The representation category is *braided* monoidal — the natural habitat of quantum groups.
- The passage $E_1 \to E_2$ via Dunn additivity: $E_2 \simeq E_1 \otimes E_1$. An E_2-chiral algebra is an E_1-algebra in E_1-algebras.

The CY condition enters through Kontsevich's identification: a d-dimensional CY structure on C determines an $\mathbb{S}^d$-framing of the Hochschild complex, hence an $\mathbb{S}^1$-action on $\mathrm{HH}_\bullet(C)$. For $d = 2$ (CY surfaces), this $\mathbb{S}^1$-action is exactly the data of an E_2-algebra structure on the cyclic homology — the braiding.

1.3 Relation to Volumes I and II

This work depends on and extends the two preceding volumes:

Volume	Provides	Used here
I (Modular Koszul Duality)	Bar-cobar machine, Θ_A, $\kappa(A)$	CY bar complex, modular trace
II (3D HT QFT)	$\mathrm{SC}^{\mathrm{ch,top}}$, PVA descent, DK bridge	E_1 sector, braided structure
III (this work)	CY $\to$ chiral functor, E_2 theory	—

1.4 CY3 combinatorics as generalized root data

The structural observation animating this work is that the combinatorics of a Calabi–Yau threefold X — its lattice, intersection form, BPS spectrum — should constitute the *generalized root datum* of a *quantum vertex chiral group* $G(X)$. For toric CY3 (via the CoHA and Drinfeld double) and for CY_2 categories (via Theorem CY-A$_2$), this object is constructed. For general CY_3 — including the prototype $K3 \times E$ — $G(X)$ is the target of a programme (Conjecture CY-A$_3$), not a constructed object.

Given a CY3 X, the generalized root datum $\mathcal{R}(X)$ consists of:

(i) A lattice $\Lambda(X)$ extracted from $K(X)$ or $H^*(X)$ with its Euler form;

(ii) Real simple roots Δ^{re} from the geometry (hyperbolic sublattice for $K3 \times E$; toric diagram for toric CY3);

(iii) Imaginary roots $\Delta^{\mathrm{im}} = \Delta^{\mathrm{im}}_0 \sqcup \Delta^{\mathrm{im}}_1$ (even and odd) with multiplicities $\mathrm{mult}(\alpha)$ from BPS counting — Donaldson–Thomas invariants for X, or equivalently the Fourier coefficients of the appropriate automorphic form;

(iv) A Weyl group $W(X)$ acting on the positive cone of $\Lambda(X)$;

(v) A Weyl vector $\rho \in \Lambda(X) \otimes \mathbb{Q}$.

Two families of examples illuminate the construction:

The $K3 \times E$ tower. For $X = (S \times E)/(\mathbb{Z}/N\mathbb{Z})$ with S a K3 surface and E an elliptic curve, the lattice is $\Lambda^{3,2} \simeq \Lambda^{1,1} \oplus \Lambda^{1,1} \oplus [2]$ of signature $(3,2)$. The hyperbolic sublattice $\Lambda^{2,1}_{II}$ with Gram matrix $\left(\begin{smallmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{smallmatrix}\right)$ provides the real roots. The root multiplicities are the Fourier coefficients $f(n,l)$ of the weak Jacobi form $\phi_{0,1}$ — the K3 elliptic genus. The resulting generalized Borcherds–Kac–Moody superalgebra $\mathfrak{g}_{\Delta_5}$ has the Igusa cusp form Δ_5 as its denominator identity.

Toric CY3 and critical CoHAs. For a toric CY3 determined by a fan Σ, the toric diagram determines a quiver Q_X. The critical cohomological Hall algebra $\mathrm{CoHA}(Q_X)$ is the positive half of an affine super Yangian $Y(\widehat{\mathfrak{g}}_{Q_X})$. For $\mathbb{C}^3$: the Jordan quiver gives $Y(\widehat{\mathfrak{gl}}_1) \simeq \mathcal{W}_{1+\infty}$. The toric fan IS the Dynkin data.

1.5 Automorphic correction as shadow tower

The passage from the "naive" Kac–Moody algebra $\mathfrak{g}$ (built from the real root Gram matrix alone) to the full generalized BKM superalgebra $\mathfrak{g}_X$ (with all imaginary roots and their multiplicities) is the *automorphic correction*. In the language of Volume I, this passage is precisely the *shadow Postnikov tower*:

Shadow tower (Vol I)	**BKM language**	**CY3 geometry**
κ (arity 2)	Weyl vector ρ contribution	Leading DT invariant
C (arity 3, cubic)	First imaginary root corrections	Genus-0 3-point function
Q (arity 4, quartic)	Higher imaginary roots	Genus-0 4-point function
Full Θ_A	Complete automorphic correction	Full BPS generating function

The denominator identity of $\mathfrak{g}_X$ — which IS the automorphic form (Δ_5 for $K3 \times E$, the DT partition function for toric CY3) — is the bar-complex Euler product of the quantum vertex chiral group. The Weyl–Kac–Borcherds character formula is the factorization structure of $B(A_X)$.

1.6 Main results

The main theorems of this work are:

- **Theorem CY-A** (CY-to-chiral functor): Construction of $\Phi\colon \mathrm{CY}_d\text{-}\mathbf{Cat} \to E_2\text{-}\mathrm{ChirAlg}$ via the factorization envelope and $\mathbb{S}^d$-framing. *Proved for* $d = 2$ (CY surfaces, where the $\mathbb{S}^2$-framing gives E_2 via Kontsevich–Vlassopoulos). *For* $d = 3$: the $d = 3$ case is a *programme*, not a theorem. The $\mathbb{S}^3$-framing gives an E_3-structure, from which E_2 is obtained by restriction; however, $\pi_1(\mathrm{Conf}_2(\mathbb{R}^3))$ is trivial, so the restriction gives a *symmetric* braiding at the topological level. The quantum group (non-symmetric) braiding for $d = 3$ is expected to arise through the Drinfeld center of E_1-monoidal categories, not through direct E_3-to-E_2 restriction. The programme is conditional on: (a) constructing the chain-level $\mathbb{S}^3$-framing compatible with BV structure, and (b) the quantization step.

- **Theorem CY-B** (E_2-chiral Koszul duality): Bar-cobar adjunction in the E_2-chiral setting. The automorphic correction of the BKM superalgebra $\mathfrak{g}_X$ is identified with the shadow tower Θ_{A_X}; the denominator identity equals the bar-complex Euler product.

- **Conjecture CY-C** (Quantum group realization): For toric CY3, $\mathrm{Rep}^{E_2}(G(X))$ should be braided monoidal equivalent to $\mathrm{Rep}(Y(\widehat{\mathfrak{g}}_{Q_X}))$; for $K3 \times E$, the $\mathrm{Sp}_4(\mathbb{Z})$-module structure on the denominator identity should recover the braided structure. The critical CoHA is the E_1-sector (positive half). The CY category $C(\mathfrak{g}, q)$ is not yet constructed in general.

- **Theorem CY-D** (Modular CY characteristic): For $d = 2$, $\kappa(\Phi(C))$ equals the CY Euler characteristic $\chi^{\mathrm{CY}}(C)$. For $K3 \times E$ ($d = 3$): the weight of Δ_5 is $5 = h^{1,1}(K3)/4 = 20/4$; this appears in the structural position of a modular characteristic, but without the chiral algebra $A_{K3\times E}$ (which is not constructed), the identification $\kappa = 5$ is an *observation* matching the Borcherds lift weight formula, not a computation from the Volume I definition of κ (AP43). The general $d = 3$ formula is conjectural.

1.7 What is proved versus what is conjectural

Proved (would survive a referee):

- The generalized root datum axioms CY1–CY7, adequate for $K3 \times E$ and toric CY3 examples.
- The E_2-chiral algebra formalism and its basic properties (definition, bar complex, braiding).
- Theorem CY-A for $d = 2$: Steps 1–3 of the functor Φ (cyclic $A_\infty \to$ Lie conformal $\to$ factorization envelope $\to E_2$ enhancement via $\mathbb{S}^2$-framing).
- The CoHA as E_1-sector for toric CY3 (via Schiffmann–Vasserot, RSYZ).
- The Drinfeld center equivalence $\mathcal{Z}(\mathrm{Rep}^{E_1}(A)) \simeq \mathrm{Rep}^{E_2}((A))$ (from Ben-Zvi–Francis–Nadler and Lurie).
- All lattice theory and BKM constructions for $K3 \times E$ (Gritsenko–Nikulin).
- The denominator identity $\Delta_5 = \Phi$ and product formula (arity 2–6 computationally verified, 837+ tests pass).

Rigorous proof sketches (completable with effort):

- The automorphic correction = shadow tower identification (parts (a)–(c)).
- The denominator = bar Euler product (conditional on CY-A for the relevant dimension).
- E_2-chiral Koszul duality (conjecture with evidence from E_1 case).

Conjectural:

- Theorem CY-A for $d = 3$: the chain-level $\mathbb{S}^3$-framing and BV-compatibility are not yet established.
- Conjecture CY-C: the CY category $C(\mathfrak{g}, q)$ is not constructed in general.
- The chiral algebra $A_{K3\times E}$ does not yet exist as a constructed object; the identification of $\kappa = 5$ with a chiral algebra modular characteristic is an observation, not a theorem.
- The Langlands = Koszul duality conjecture.
- All physics conjectures in Part VI.

1.8 Guide for the reader

Part I establishes the categorical foundations: CY categories, cyclic A_∞-structures, and the Hochschild calculus. Part II develops the E_1 and E_2 chiral theories. Part III constructs the bridge functor Φ and proves Theorems CY-A through CY-D, with the identification of automorphic correction and shadow tower as the central result. Part IV studies quantum groups and their braided factorization structure, including the Drinfeld center as the bulk algebra. Part V works through the standard landscape: Fukaya categories, derived categories, matrix factorizations (ADE singularities and W-algebras), and quantum group representations (Kazhdan–Lusztig, affine Yangians, critical CoHAs). Part VI connects back to Volumes I and II and to the geometric Langlands programme.

Chapter 2

Calabi–Yau Categories

2.1 Smooth and proper dg categories

Definition 2.1 *(Smooth dg category).* A dg category C over k is *smooth* if the diagonal bimodule $C \in C^{\mathrm{op}} \otimes C\text{-}\mathrm{Mod}$ is a perfect bimodule (compact in the derived category of bimodules).

Definition 2.2 *(Proper dg category).* A dg category C is *proper* if for all objects $X, Y \in C$, the Hom-complex $\mathrm{Hom}_C(X, Y)$ has finite-dimensional total cohomology.

2.2 The Calabi–Yau condition

Definition 2.3 *(Calabi–Yau category, after Kontsevich).* A smooth dg category C is *Calabi–Yau of dimension d* if there is a quasi-isomorphism of C-bimodules

$$C \xrightarrow{\sim} C^{\vee}[d]$$

where $C^{\vee} = \mathrm{RHom}_{C^{\mathrm{op}} \otimes C}(C, C^{\mathrm{op}} \otimes C)$ is the inverse dualizing bimodule. Equivalently, there exists a non-degenerate cyclic pairing

$$\langle \cdot, \cdot \rangle \colon \mathrm{HH}_{\bullet}(C) \otimes \mathrm{HH}_{\bullet}(C) \longrightarrow k[-d].$$

Remark 2.4. The CY condition combines two ingredients: smoothness (the bimodule C is compact) and the Serre functor being a shift ($S_C \simeq [d]$). A smooth and proper CY category is *saturated* in the sense of Kontsevich–Soibelman.

2.3 The CY trace

Definition 2.5 *(Non-degenerate trace).* A *d-dimensional CY structure* on a smooth dg category C is a class

$$[\sigma] \in \mathrm{HC}^{-}_{d}(C)$$

in the negative cyclic homology, whose image under the canonical map $\mathrm{HC}^{-} \to \mathrm{HH}$ is a non-degenerate trace $\mathrm{Tr} \colon \mathrm{HH}_d(C) \to k$.

2.4 Standard examples

Example 2.6 *(Fukaya category).* For a compact symplectic manifold (M, ω) with $c_1(M) = 0$, the Fukaya category $\mathrm{Fuk}(M)$ is CY of dimension $\dim_{\mathbb{R}}(M)/2 = n$. The CY trace arises from the cyclic structure on the A_∞-algebra of Lagrangian Floer cochains.

Example 2.7 *(Derived category of a CY manifold).* For a smooth projective CY manifold X of dimension n (i.e., $\omega_X \simeq \mathcal{O}_X$, $H^i(X, \mathcal{O}_X) = 0$ for $0 < i < n$), the bounded derived category $D^b(\mathrm{Coh}(X))$ is CY of dimension n. The CY trace is induced by the Serre duality pairing.

Example 2.8 *(Matrix factorizations).* For $W : \mathbb{A}^n \to \mathbb{A}^1$ a polynomial with isolated singularity, the category $\mathrm{MF}(W)$ of matrix factorizations is CY of dimension $n - 2$ (Dyckerhoff; Orlov). For the ADE singularities in two variables ($n = 2$), $\mathrm{MF}(W)$ is CY_0 (semisimple, with the structure of a Frobenius algebra). For threefold singularities ($n = 4$, e.g. $W = x_1^2 + \cdots + x_4^2$), $\mathrm{MF}(W)$ is CY_2.

Chapter 3

Cyclic A_∞-Structures

3.1 A_∞-categories and the bar complex

The cyclic bar complex is the primary invariant of a CY category. We review the A_∞-formalism and its cyclic refinement, preparing for the bridge to chiral algebras in Part III.

Definition 3.1 *(A_∞-category).* An A_∞-category C over k consists of:

(i) A set of objects $\mathrm{Ob}(C)$;

(ii) For each pair of objects X, Y, a graded k-module $\mathrm{Hom}_C(X, Y)$;

(iii) Higher composition maps $\mu_n \colon \mathrm{Hom}(X_0, X_1) \otimes \cdots \otimes \mathrm{Hom}(X_{n-1}, X_n) \to \mathrm{Hom}(X_0, X_n)[2-n]$ for $n \geq 1$, satisfying the A_∞-relations:

$$\sum_{\substack{p+q=n+1 \\ 1 \leq i \leq p}} (-1)^{\dagger} \, \mu_p(a_1, \ldots, a_{i-1}, \mu_q(a_i, \ldots, a_{i+q-1}), a_{i+q}, \ldots, a_n) = 0$$

where the sign $(-1)^{\dagger}$ follows the Koszul rule (see Appendix A).

3.2 Cyclic structures

Definition 3.2 *(Cyclic A_∞-algebra).* A *cyclic A_∞-algebra of dimension d* is an A_∞-algebra $(A, \{\mu_n\}_{n\geq 1})$ equipped with a non-degenerate pairing $\langle \cdot, \cdot \rangle \colon A \otimes A \to k[-d]$ satisfying the cyclic invariance condition

$$\langle \mu_n(a_1, \ldots, a_n), a_0 \rangle = \pm \langle \mu_n(a_0, a_1, \ldots, a_{n-1}), a_n \rangle$$

for all $n \geq 1$, where the sign is determined by the Koszul rule.

Remark 3.3. The cyclic invariance condition is the chain-level precursor of the chiral algebra trace. The passage from cyclic A_∞ to chiral algebra requires promoting the cyclic pairing to a factorization structure on a curve — this is the content of Part III.

3.3 The cyclic bar complex

Construction 3.4 *(Cyclic bar complex).* Given a cyclic A_∞-algebra $(A, \langle\cdot,\cdot\rangle)$ of dimension d, the *cyclic bar complex* is

$$\mathrm{CC}_\bullet(A) = \bigoplus_{n\geq 0} (A[1])^{\otimes n}/(\text{cyclic})$$

with differential induced by the A_∞-operations and the cyclic quotient enforced by the pairing. The S^1-action by cyclic rotation on $\mathrm{CC}_\bullet(A)$ endows the cyclic homology $\mathrm{HC}_\bullet(A)$ with its defining structure.

3.4 The $\mathbb{S}^d$-framing

A d-dimensional CY structure enhances the S^1-action to an $\mathbb{S}^d$-action on the Hochschild complex. This is the key datum: for $d = 2$, the $\mathbb{S}^2$-framing provides the E_2-algebra structure that will underpin the braided monoidal/quantum group side of the story.

Chapter 4

Hochschild Calculus for CY Categories

4.1 Hochschild homology and cohomology

4.2 The Hochschild-to-cyclic spectral sequence

4.3 CY Hochschild duality

THEOREM 4.1 *(CY Hochschild duality).* [PE] For a smooth CY category C of dimension d, there is a canonical isomorphism

$$\mathrm{HH}^{\bullet}(C) \simeq \mathrm{HH}_{\bullet+d}(C)$$

identifying Hochschild cohomology with (shifted) Hochschild homology. Under this isomorphism, the Gerstenhaber bracket on $\mathrm{HH}^{\bullet}$ corresponds to the BV operator (divergence operator) on $\mathrm{HH}_{\bullet}$; the Connes B-operator on $\mathrm{HH}_{\bullet}$ corresponds to the de Rham differential under the HKR decomposition.

Remark 4.2. The isomorphism $\mathrm{HH}^{\bullet} \simeq \mathrm{HH}_{\bullet+d}$ is sometimes called "CY duality" or "Poincaré duality for Hochschild (co)homology." It follows from the smoothness of C (the diagonal bimodule is compact) combined with the CY condition ($C \simeq C^{\vee}[d]$). See Kontsevich (2006 ICM), Keller (2006), Ginzburg (2006 arXiv:0612139).

4.4 Categorical Hodge theory

Part II

E_1 and E_2 Chiral Theories

Chapter 5

E_1-Chiral Algebras

The E_1-chiral theory was developed in Volume II (Part VII) as the ordered sector of the Swiss-cheese algebra. Here we reformulate it as the foundation for the E_2 extension.

5.1 E_1-algebras and associative factorization

An E_1-algebra is an algebra over the little intervals operad. In the chiral setting, an E_1-chiral algebra factorizes associatively along a real line $\mathbb{R}$ while maintaining holomorphic structure along a transverse curve.

Definition 5.1 *(E_1-chiral algebra).* An *E_1-chiral algebra* on $X \times \mathbb{R}$ is a factorization algebra $\mathcal{A}$ on $\mathrm{Ran}(X) \times \mathrm{Ran}(\mathbb{R})$ such that:

(i) The X-direction factorization is holomorphic (chiral);

(ii) The $\mathbb{R}$-direction factorization is E_1 (associative).

The representation category $\mathrm{Rep}^{E_1}(\mathcal{A})$ is a monoidal category.

5.2 The bar complex as E_1-chiral algebra

Proposition 5.2. For a chiral algebra A on X, the bar complex $B(A)$ carries a natural E_1-chiral structure: the bar differential encodes the X-direction (holomorphic OPE residues) and the bar coproduct encodes the $\mathbb{R}$-direction (ordered deconcatenation).

This is the content of the Swiss-cheese algebra structure from Volume II.

5.3 Monoidal representation categories

5.4 The E_1-Koszul duality

Theorem 5.3 *(E_1-chiral Koszul duality, Volume II).* [PE] For an E_1-chiral algebra $\mathcal{A}$, the bar-cobar adjunction of Volume I refines to an E_1-equivariant adjunction. On the representation-category level, this gives a monoidal equivalence

$$\mathrm{Rep}^{E_1}(A) \simeq \mathrm{Rep}^{E_1}(A^!)^{\mathrm{rev}}$$

between the monoidal representation category of A and the reverse monoidal category of $A^!$.

Chapter 6

E_2-Chiral Algebras

This is the central innovation of the present work. An E_2-chiral algebra factorizes along two holomorphic directions, and its representation category is *braided* monoidal — the natural algebraic structure underlying quantum groups.

6.1 The E_2 operad and braided monoidal categories

Definition 6.1 *(E_2 operad).* The E_2 operad (little 2-disks operad) has $E_2(n) = \mathrm{Conf}_n(\mathbb{D}^2)$, the configuration space of n non-overlapping disks in the unit disk. An E_2-algebra is an algebra over this operad. By Dunn additivity, $E_2 \simeq E_1 \otimes E_1$: an E_2-algebra is an E_1-algebra in E_1-algebras.

THEOREM 6.2 *(Lurie).* The category of E_2-algebras in a symmetric monoidal ∞-category C is equivalent to the category of braided monoidal objects in C.

6.2 E_2-chiral algebras: definition and first properties

Definition 6.3 *(E_2-chiral algebra).* An *E_2-chiral algebra* on $X \times Y$ (where X, Y are smooth curves) is a factorization algebra $\mathcal{A}$ on $\mathrm{Ran}(X) \times \mathrm{Ran}(Y)$ such that:

(i) The X-direction factorization is chiral (holomorphic);

(ii) The Y-direction factorization is chiral (holomorphic);

(iii) The combined structure is compatible with the E_2-operad action via the identification $\overline{\mathrm{FM}}_n(\mathbb{C}) \simeq E_2(n)$.

The representation category $\mathrm{Rep}^{E_2}(\mathcal{A})$ is a *braided monoidal* category.

6.3 Kontsevich formality and the E_2-Gerstenhaber structure

Kontsevich's formality theorem $E_2 \xrightarrow{\sim} \mathrm{Ger}$ (over $\mathbb{Q}$) identifies E_2-algebras with Gerstenhaber algebras up to homotopy. In the chiral context, this gives:

PROPOSITION 6.4. The Hochschild cohomology $\mathrm{HH}^\bullet(A)$ of a chiral algebra A carries a natural E_2-algebra structure. Via Kontsevich formality, this is equivalent to the Gerstenhaber bracket + cup product structure.

6.4 The E_2 bar complex

Construction 6.5 *(E_2 bar complex).* For an E_2-chiral algebra $\mathcal{A}$ on $X \times Y$, the E_2 *bar complex* $B_{E_2}(\mathcal{A})$ is a factorization coalgebra on $\mathrm{Ran}(X) \times \mathrm{Ran}(Y)$ with:

- Two commuting differentials: d_X from X-direction OPE residues, d_Y from Y-direction OPE residues;
- Two commuting coproducts: Δ_X from X-direction deconcatenation, Δ_Y from Y-direction deconcatenation;
- The braiding: the transposition of the two E_1-structures gives the R-matrix.

6.5 E_2-chiral Koszul duality

CONJECTURE 6.6 *(E_2-chiral Koszul duality).* [CJ] For an E_2-chiral algebra $\mathcal{A}$, the bar-cobar adjunction of Volume I extends to an E_2-equivariant adjunction. On representation categories, this gives a braided monoidal equivalence

$$\mathrm{Rep}^{E_2}(A) \simeq \mathrm{Rep}^{E_2}(A^!)^{\mathrm{rev}}$$

where rev denotes the reverse braiding.

Chapter 7

E_n-Factorization and Higher Chiral Structure

7.1 Dunn additivity and the E_n hierarchy

7.2 Factorization algebras on $\mathbb{C}^n$

7.3 The E_n-chiral bar complex

7.4 Relation to topological field theory

Part III

The Bridge: Calabi–Yau Categories as Quantum Chiral Algebras

Chapter 8

From CY Categories to Chiral Algebras

This chapter constructs the bridge functor Φ from CY categories to quantum chiral algebras.

8.1 The cyclic-to-chiral passage

A CY category C of dimension d carries a cyclic A_∞-structure (Chapter 3). The cyclic bar complex $\mathrm{CC}_\bullet(C)$ with its $\mathbb{S}^d$-framing is the primary invariant. The passage to chiral algebras proceeds by:

Step 1. **Cyclic $A_\infty \to$ Lie conformal algebra.** The Hochschild cohomology $\mathrm{HH}^\bullet(C)$, with its Gerstenhaber bracket, determines a graded Lie algebra. The CY pairing promotes this to a Lie conformal algebra $\mathfrak{L}_C$ (the "current algebra" of C).

Step 2. **Lie conformal algebra $\to$ factorization envelope.** The factorization envelope construction (Nishinaka–Vicedo) produces a factorization algebra $\mathrm{Fact}_X(\mathfrak{L}_C)$ on any smooth curve X.

Step 3. **E_2 enhancement.** When $d = 2$, the $\mathbb{S}^2$-framing of $\mathrm{HH}_\bullet(C)$ provides an E_2-algebra structure. This enhances $\mathrm{Fact}_X(\mathfrak{L}_C)$ to an E_2-chiral algebra.

Step 4. **Quantization.** The quantum chiral algebra A_C is the quantization of $\mathrm{Fact}_X(\mathfrak{L}_C)$ determined by the CY trace.

8.2 The CY-to-chiral functor

Theorem 8.1 *(CY-to-chiral functor — Theorem CY-A$_2$, $d = 2$).* There exists a functor

$$\Phi\colon \mathrm{CY}_2\text{-}\mathbf{Cat} \longrightarrow E_2\text{-}\mathrm{ChirAlg}$$

from smooth proper CY categories of dimension $d = 2$ to E_2-chiral algebras, such that:

(i) $\Phi(C)$ is a chiral algebra whose generating fields are in bijection with $\mathrm{HH}^{\bullet+1}(C)$;

(ii) The bar complex $B(\Phi(C))$ is quasi-isomorphic to the cyclic bar complex $\mathrm{CC}_\bullet(C)$ as a factorization coalgebra;

(iii) The modular characteristic $\kappa(\Phi(C))$ equals the CY Euler characteristic.

The E_2-enhancement in Step 3 uses the $\mathbb{S}^2$-framing of $\mathrm{HH}_\bullet(C)$ (Kontsevich–Vlassopoulos).

Conjecture 8.2 *(CY-to-chiral programme — CY-A_3, $d = 3$).* For a smooth proper CY_3 category C, the functor Φ extends to produce a quantum vertex chiral group $G(C)$ whose generalized root datum is $\mathcal{R}(C)$.

The $\mathbb{S}^3$-framing gives an E_3-structure on $\mathrm{HH}_\bullet(C)$. The E_3-structure restricts to E_2 via Dunn additivity, but this restricted braiding is *symmetric* at the topological level, since $\pi_1(\mathrm{Conf}_2(\mathbb{R}^3))$ is trivial. The quantum group (non-symmetric) braiding for $d = 3$ does NOT arise from the E_3-to-E_2 restriction; it is expected to arise through the *Drinfeld center* of the E_1-monoidal representation category (see Chapter 13).

This programme is conditional on: (a) constructing the chain-level $\mathbb{S}^3$-framing compatible with the BV structure, and (b) establishing the quantization step (Step 4) for $d = 3$.

8.3 Compatibility with Koszul duality

8.4 The CY modular characteristic

Chapter 9

Quantum Chiral Algebras

9.1 Definition and structure

A *quantum chiral algebra* is an E_2-chiral algebra A (Definition 6.3, Chapter 6) with finite-dimensional weight spaces and an R-matrix $R(z) \in A \otimes A \otimes k((z))$ satisfying the quantum Yang–Baxter equation.

Conjecture 9.1 *(Quantum vertex chiral group — target specification).* [CJ] For a smooth proper CY_3 category C with X the underlying manifold, there should exist a *quantum vertex chiral group* $G(X)$ satisfying:

(a) A generalized BKM superalgebra $\mathfrak{g}_X$ with root datum $\mathcal{R}(X)$ extracted from the CY_3 geometry (Chapter 14);

(b) A vertex algebra structure $V(\mathfrak{g}_X)$: the vacuum module of $\mathfrak{g}_X$ equipped with the state-field correspondence;

(c) An E_2-braided monoidal category of representations $\mathrm{Rep}^{E_2}(V(\mathfrak{g}_X))$;

(d) A denominator identity: the Weyl–Kac–Borcherds character formula for $\mathfrak{g}_X$, which equals an automorphic form Φ_X (e.g., Δ_5 for $K3 \times E$).

Warning 9.2 (Status of $G(X)$). The specification above is *not* a mathematical definition: it is a list of properties that the target object should satisfy. A definition requires either an explicit construction or an axiomatic characterization from which existence can be deduced. Neither is available for general CY_3. What IS established:

- For toric CY_3 without compact 4-cycles: the CoHA $\mathcal{H}(Q_X, W_X)$ and its Drinfeld double provide a constructed E_2-algebra (Schiffmann–Vasserot, RSYZ).
- For $d = 2$: the functor Φ of Theorem CY-A$_2$ constructs A_C from a CY_2 category.
- For $K3 \times E$ ($d = 3$): the BKM superalgebra $\mathfrak{g}_{\Delta_5}$ and its root datum exist (Gritsenko–Nikulin), but the chiral algebra $A_{K3\times E} = \Phi(D^b(\mathrm{Coh}(X)))$ is *not* constructed. The identification $\kappa = 5$ is an observation matching the Borcherds lift weight, not a computation from Vol I's definition of κ.
- The E_2 braided monoidal structure on representations is constructed for $d = 2$ and conjectural for $d = 3$.

Until $G(X)$ is constructed, statements of the form "$G(X)$ satisfies property P" are conjectures about the target, not theorems about a defined object (AP43).

The key structural equation: the E_2-structure on A determines a universal R-matrix

$$R(z) \in A \otimes A \otimes k((z))$$

satisfying the quantum Yang–Baxter equation. This R-matrix is the E_2 analogue of the collision r-matrix $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_A)$ from Volume I (whose pole orders are one less than the OPE poles, due to the $d\log$ extraction; see Vol I, AP19).

9.2 The CY R-matrix

Construction 9.3 *(CY R-matrix).* Given a CY category $\mathcal{C}$ of dimension 2 and its quantum chiral algebra $A_{\mathcal{C}} = \Phi(\mathcal{C})$, the CY R-matrix is

$$R_{\mathcal{C}}(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{A_{\mathcal{C}}})$$

where $\Theta_{A_{\mathcal{C}}}$ is the universal MC element from Volume I. The pole structure of $R_{\mathcal{C}}(z)$ encodes the CY shadow depth classification.

9.3 Quantum chiral algebras from quantum groups

9.4 The quantum chiral Koszul dual

Chapter 10

The Modular Trace

10.1 CY trace as modular characteristic

The CY trace $\mathrm{Tr}\colon \mathrm{HH}_d(C) \to k$ determines the modular characteristic $\kappa(A_C)$ of the associated quantum chiral algebra.

THEOREM 10.1 *(CY modular characteristic — Theorem CY-D).* [PH] For a CY category C of dimension $d = 2$ with quantum chiral algebra $A_C = \Phi(C)$:

(i) The modular characteristic $\kappa(A_C) = \chi^{\mathrm{CY}}(C)$, the CY Euler characteristic;

(ii) The genus-g obstruction satisfies $\mathrm{obs}_g(A_C) = \kappa(A_C) \cdot \lambda_g$ on the uniform-weight lane; unconditionally at genus 1 for all families (Vol I, Theorem D); at genus $g \geq 2$ for multi-weight algebras, this is open (Vol I, op:multi-generator-universality);

(iii) The shadow tower of A_C encodes the higher-genus CY invariants of C.

10.2 Genus expansion and Gromov–Witten invariants

10.3 The CY shadow tower

10.4 Complementarity for CY pairs

Part IV

Quantum Groups and Braided Monoidal Structure

Chapter 11

Quantum Groups: Foundations

Chapter 12

Braided Factorization

12.1 Braided monoidal categories and E_2-algebras

12.2 Factorization homology with E_2-coefficients

12.3 The braided bar-cobar adjunction

THEOREM 12.1 *(E_2-bar-cobar — Theorem CY-B).* For an E_2-chiral algebra $\mathcal{A}$, the bar-cobar adjunction

$$B_{E_2} : E_2\text{-ChirAlg} \rightleftarrows E_2\text{-ChirCoalg}\Omega_{E_2}$$

satisfies:

(i) The E_2 bar complex $B_{E_2}(\mathcal{A})$ is a factorization E_2-coalgebra;

(ii) Cobar-bar inversion: $\Omega_{E_2}(B_{E_2}(\mathcal{A})) \simeq \mathcal{A}$ on the Koszul locus;

(iii) The CY trace manifests as curvature: at genus $g \geq 1$, $d^2 = \kappa \cdot \omega_g$.

Remark 12.2 (Status of Theorem CY-B). Items (i)–(iii) follow from the Volume I bar-cobar machine (Theorems A and B) applied to each E_1-direction separately, together with the compatibility of the two E_1-structures. The full E_2-equivariant refinement — in particular, that the R-matrix intertwines the two bar-cobar adjunctions coherently — is a rigorous proof sketch (completable with effort), not a fully written proof. The identification of the automorphic correction with the shadow tower (claimed in Theorem CY-B for $K3 \times E$) is conditional on Conjecture CY-A$_3$ for the existence of $A_{K3\times E}$.

12.4 Quantum group structure from braided Koszul duality

Chapter 13

The Drinfeld Center and Bulk Algebras

13.1 The Drinfeld center of a monoidal category

Definition 13.1 *(Drinfeld center).* For a monoidal category C, the *Drinfeld center* $\mathcal{Z}(C)$ is the category of pairs $(X, \beta_{X,-})$ where $X \in C$ and $\beta_{X,-} : X \otimes (-) \xrightarrow{\sim} (-) \otimes X$ is a half-braiding satisfying the hexagon axiom. The Drinfeld center is always braided monoidal.

13.2 Drinfeld center as chiral derived center

Proposition 13.2. For an E_1-chiral algebra A with monoidal representation category $C = \mathrm{Rep}^{E_1}(A)$, the Drinfeld center $\mathcal{Z}(C)$ is equivalent to the representation category of the chiral derived center:

$$\mathcal{Z}(\mathrm{Rep}^{E_1}(A)) \simeq \mathrm{Rep}^{E_2}(Z^{\mathrm{der}}_{\mathrm{ch}}(A)).$$

This is the categorical incarnation of the bulk-boundary correspondence: the E_1 (boundary) theory determines an E_2 (bulk) theory via the Drinfeld center.

13.3 CY enhancement of the Drinfeld center

13.4 Quantum group reconstruction from the center

Part V

The Standard Landscape

Chapter 14

The $K3 \times E$ Tower and the Igusa Cusp Form

This chapter develops the prototypical example of a quantum vertex chiral group: the generalized BKM superalgebra $\mathfrak{g}_{\Delta_5}$ attached to the CY3 family $X = (S \times E)/(\mathbb{Z}/N\mathbb{Z})$, whose denominator identity is the Igusa cusp form Δ_5.

14.1 The CY3 geometry

Let (E, e_0) be an elliptic curve with an N-torsion point and S a K3 surface with elliptic fibration $\pi\colon S \to \mathbb{P}^1$ admitting sections $s_1, s_2\colon \mathbb{P}^1 \to S$ with s_2 of order N relative to s_1. The product $S \times E$ admits a free $\mathbb{Z}/N\mathbb{Z}$-action

$$(s, e) \longmapsto (s + s_2(\pi(s)), e + e_0),$$

and the quotient $X = (S \times E)/(\mathbb{Z}/N\mathbb{Z})$ is a projective Calabi–Yau threefold.

Definition 14.1 *(The DT zeta function).* Let F be the fiber class of π and $\beta_h = \frac{1}{N}(s_1 + \cdots + s_N + hF)$ for $h \geq 0$. The DT zeta function of X is

$$Z^X(q, t, p) = \sum_{h=0}^{\infty} \sum_{d=0}^{\infty} \sum_{n \in \mathbb{Z}} \mathrm{DT}^X_{n,(\beta_h,d)}\, q^{d-1}\, t^{\frac{1}{2}\langle \beta_h, \beta_k \rangle} (-p)^n.$$

Theorem 14.2 *(Oberdieck–Pixton).* For $X = S \times E$ of order $N = 1$:

$$Z^X(q, t, p) = \frac{C}{(\Delta_5)^2}$$

for a constant C, where Δ_5 is the Igusa cusp form of weight 5 for $\mathrm{Sp}_4(\mathbb{Z})$.

The elliptic genus of K3 — the weak Jacobi form $\phi_{0,1}$ of weight 0 and index 1 — governs the root multiplicities of the BKM superalgebra.

14.2 The lattice $\Lambda^{3,2}$ and the isomorphism $\mathrm{Sp}_4 \simeq \mathrm{SO}_+$

Consider the rank-4 free $\mathbb{Z}$-module $\Lambda^4 = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2 \oplus \mathbb{Z}e_3 \oplus \mathbb{Z}e_4$. The exterior square $\bigwedge^2\colon \mathrm{SL}_4(\mathbb{Z}) \to \mathrm{GL}(\bigwedge^2 \Lambda^4)$ and the Pfaffian scalar product $(u, v) = u \wedge v/(e_1 \wedge e_2 \wedge e_3 \wedge e_4)$ give a signature $(3, 3)$

form. The lattice

$$\Lambda^{3,2} = (e_1 \wedge e_3 + e_2 \wedge e_4)^{\perp} \subset \bigwedge^2 \Lambda^4$$

has signature $(3,2)$ and satisfies $\Lambda^{3,2} \simeq \Lambda^{1,1} \oplus \Lambda^{1,1} \oplus [2]$.

LEMMA 14.3. The correspondence $\bigwedge^2$ defines an isomorphism

$$\bigwedge^2 : \mathrm{Sp}_4(\mathbb{Z})/\{\pm I_4\} \xrightarrow{\sim} \mathrm{O}(\Lambda^{3,2})_+/\{\pm I_5\}.$$

In the basis $(f_1, f_2, f_3, f_{-2}, f_{-1})$ with $f_1 = e_1 \wedge e_2$, $f_2 = e_2 \wedge e_3$, $f_3 = e_1 \wedge e_3 - e_2 \wedge e_4$, $f_{-2} = e_4 \wedge e_1$, $f_{-1} = e_4 \wedge e_3$, the orthogonal group $\mathrm{O}_{\mathbb{R}}(\Lambda^{3,2})_+$ acts on the type IV domain $\mathbb{H}_+^{IV}$, which identifies with the Siegel upper half-space $\mathbb{H}_2$.

14.3 The hyperbolic sublattice and reflection groups

Fix the primitive hyperbolic sublattice

$$\Lambda^{2,1} = \Lambda^{1,1} \oplus [2] \simeq \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2} \subset \Lambda^{3,2}.$$

The sublattice $\Lambda^{2,1}_{II}$ generated by elements of type II (those δ with $(\delta, \Lambda^{2,1}) = 2\mathbb{Z}$) has three real simple roots:

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3,$$

with Gram matrix

$$(\delta_i, \delta_j) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

The fundamental polyhedron $\mathcal{P}_{II}$ has $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$, and

$$\mathrm{O}(\Lambda^{2,1})_+ \simeq W^{(2)}(\Lambda^{2,1}_{II}) \rtimes \mathrm{Aut}(\mathcal{P}_{II}).$$

Definition 14.4 *(Weyl vector).* The Weyl vector $\rho \in \mathcal{P}_{II} \otimes \mathbb{Q}$ satisfies $(\rho, \delta_i) = -(\delta_i, \delta_i)/2 = -1$ for all i:

$$\rho = \tfrac{1}{2}\delta_1 + \tfrac{1}{2}\delta_2 + \tfrac{1}{2}\delta_3 = f_2 - \tfrac{1}{2}f_3 + f_{-2}.$$

14.4 The generalized BKM superalgebra $\mathfrak{g}_{\Delta_5}$

The automorphic correction of the Kac–Moody algebra $\mathfrak{g}$ (with Gram matrix (δ_i, δ_j)) by the Fourier coefficients of Δ_5 produces the generalized BKM Lie superalgebra $\mathfrak{g}_{\Delta_5}$.

Construction 14.5 *(Root system of $\mathfrak{g}_{\Delta_5}$).* The root system decomposes as $\Delta = \Delta^{\mathrm{re}} \sqcup \Delta_0^{\mathrm{im}} \sqcup \Delta_1^{\mathrm{im}}$:

- **Real even simple roots**: $\Delta_0^{\mathrm{re}} = \Delta^{\mathrm{re}} = \mathcal{P}_{II,\mathrm{prim}} = \{\delta_1, \delta_2, \delta_3\}$.
- **Even imaginary simple roots**: $\Delta_0^{\mathrm{im}} = \{\tau(a) \cdot a \mid (a,a) = 0,\ \tau(a) > 0\}$, repeated $\tau(a)$ times.
- **Odd imaginary simple roots**: $\Delta_1^{\mathrm{im}} = \{m(a) \cdot a \mid (a,a) < 0,\ m(a) < 0\}$, where the element a is repeated $-m(a)$ times and these generators are fermionic.

Here $m(a) = -\frac{1}{64} f(n,l,m)$ for $a \in \Lambda_{II}^{2,1} \cap \mathbb{R}_{\geq 0}\mathcal{P}_{II}$ corresponding to the Fourier coefficient $f(n,l,m)$ of Δ_5.

The superalgebra $\mathfrak{g}_{\Delta_5}$ has the triangular decomposition

$$\mathfrak{g}_{\Delta_5} = \left(\bigoplus_{\alpha \in \widetilde{C(\Lambda_{II}^{2,1})_+}} \mathfrak{g}_\alpha \right) \oplus (\Lambda_{II}^{2,1} \otimes \mathbb{R}) \oplus \left(\bigoplus_{\alpha \in -\widetilde{C(\Lambda_{II}^{2,1})_+}} \mathfrak{g}_\alpha \right)$$

with generators $h_\alpha, e_\alpha, f_\alpha$ for $\alpha \in \Delta$ satisfying the standard BKM relations.

14.5 The denominator identity: $\Delta_5 = \Phi$

THEOREM 14.6 *(Denominator identity).* The Weyl–Kac–Borcherds character formula applied to the trivial one-dimensional representation of $\mathfrak{g}_{\Delta_5}$ gives

$$\frac{1}{64}\Delta_5(2Z) = \Phi(z)$$

where Φ is the denominator function

$$\Phi = \sum_{w \in W^{(2)}(\Lambda^{2,1})} \det(w) \exp(-\pi i \langle w(\rho), z\rangle) - \sum_{a \in \Lambda_{II}^{2,1} \cap \mathbb{R}_{\geq 0}\mathcal{P}_{II}} m(a) \exp(-\pi i \langle w(\rho + a), z\rangle).$$

Equivalently, in product form:

$$\Phi = \exp(-2\pi i \langle \rho, z\rangle) \prod_{\alpha \in \Delta_+} (1 - \exp(-2\pi i \langle \alpha, z\rangle))^{\mathrm{mult}\,\alpha}.$$

THEOREM 14.7 *(Product formula for Δ_5).*

$$\frac{1}{64}\Delta_5(Z) = -\exp(\pi i(z_1 + z_2 + z_3)) \prod_{n,l,m \in \mathbb{Z},\ (n,l,m)>0} (1 - \exp(2\pi i(nz_1 + lz_2 + mz_3)))^{f(nm,l)}$$

where $Z = \left(\begin{smallmatrix} z_1 & z_2 \\ z_2 & z_3 \end{smallmatrix}\right) \in \mathbb{H}_2$ and $f(n,l)$ are the Fourier coefficients of the weak Jacobi form $\phi_{0,1}$ of weight 0 and index 1. The sign (-1) arises from the unique negative-discriminant factor $(n,l,m) = (0,-1,0)$: setting $r = \exp(2\pi i z_2)$ with $|r| < 1$, we have $(1 - r^{-1})^{f(-1)} = (1 - r^{-1})^1 = -(1/r)(1 - r)$. Absorbing this factor, the sign-free form reads

$$\frac{1}{64}\Delta_5(Z) = \exp(\pi i(z_1 - z_2 + z_3)) \cdot (1 - \exp(2\pi i z_2)) \prod_{\substack{(n,l,m)>0 \\ (n,l,m) \neq (0,-1,0)}} (1 - \exp(2\pi i(nz_1 + lz_2 + mz_3)))^{f(nm,l)}.$$

[PH]

14.6 The weak Jacobi form $\phi_{0,1}$ and root multiplicities

The weak Jacobi form $\phi_{0,1} = \phi_{12,1}/\delta_{12}$ (where $\delta_{12} = q\prod_{n\geq 1}(1 - q^n)^{24}$ is the weight-12 cusp form) is the K3 elliptic genus. Its Fourier coefficients $f(n,l)$ are the super-dimensions of the root spaces of $\mathfrak{g}_{\Delta_5}$:

$$\mathrm{mult}\,\alpha = f(nm, l) \quad \text{for } \alpha = (n,l,m) \in \Delta_+.$$

14.7 The quantum vertex chiral group $G(K3 \times E)$

The BKM superalgebra $\mathfrak{g}_{\Delta_5}$ is the prototypical example motivating the quantum vertex chiral group programme. In the language of Volumes I–III:

- **(Theorem.)** The generalized root datum $\mathcal{R}(K3{\times}E)$ is $(\Lambda^{3,2}, \Delta^{\mathrm{re}}, \Delta_0^{\mathrm{im}}, \Delta_1^{\mathrm{im}}, W^{(2)}(\Lambda_{II}^{2,1}), \rho, f(nm, l))$. This is a mathematical fact (Gritsenko–Nikulin).
- **(Conjecture.)** There exists a chiral algebra $A_{K3\times E} = \Phi(D^b(\mathrm{Coh}(X)))$ whose bar complex $B(A)$ encodes the product formula for Δ_5. *Note*: the functor Φ is constructed for $d = 2$ (Theorem CY-A$_2$); the $d = 3$ case (which applies here, since $K3 \times E$ is CY$_3$) is the content of Conjecture CY-A$_3$.
- **(Observation/Conjecture.)** The number $5 = \mathrm{wt}(\Delta_5) = h^{1,1}(K3)/4$ appears in the structural position of a modular characteristic: $\kappa = 5$. Without the chiral algebra $A_{K3\times E}$, this identification is an observation matching the Borcherds lift weight formula, not a computation from the Volume I definition of κ.
- **(Theorem at the formal product level.)** The automorphic correction $\mathfrak{g} \rightsquigarrow \mathfrak{g}_{\Delta_5}$ matches the shadow tower structure at the level of the formal Euler product: the arity-r projection captures imaginary roots of BPS charge $\leq r$. This is computationally verified at arities 2–6. *Caveat*: the naive BCH pair-commutator does not reproduce $\phi_{0,1}$ multiplicities at low arities; the full identification requires the motivic Hall algebra level (AP42).
- **(Conjectural.)** The E_2-structure should come from the $\mathrm{Sp}_4(\mathbb{Z})$-action on $\mathbb{H}_2$, connecting genus-2 modular structure to braided monoidal structure via the modular functor. Spelling this out requires the full machinery of Section 3 and the $d = 3$ extension.

CONJECTURE 14.8 *(Eight quantum vertex chiral groups).* Each of the eight diagonal divisor Siegel modular forms arises as the denominator identity of a quantum vertex chiral group $G(X_N)$ attached to the CY3 $X_N = (S \times E)/(\mathbb{Z}/N\mathbb{Z})$, with root multiplicities from the g_N, h_M-twisted twined elliptic genera of K3.

Chapter 15

Toric CY3 and Critical CoHAs

This chapter develops the second main class of examples: toric Calabi–Yau threefolds, whose critical cohomological Hall algebras provide the positive half of affine super Yangians.

15.1 Toric CY3 threefolds and their quivers

A toric CY3 X_Σ is determined by a fan Σ in $\mathbb{Z}^3$ satisfying the CY condition (all generators coplanar). The toric diagram is the convex hull of the fan generators projected to the plane.

Example 15.1 *($\mathbb{C}^3$).* The simplest toric CY3. Toric diagram: a single triangle. Quiver: the Jordan quiver (one vertex, one loop).

Example 15.2 *(Resolved conifold).* Toric diagram: two triangles sharing an edge. Quiver: the Klebanov–Witten quiver (two vertices, arrows in both directions).

15.2 The critical cohomological Hall algebra

Definition 15.3 *(Critical CoHA).* For a quiver with potential (Q, W) coming from a toric CY3 X, the *critical CoHA* is the $\mathbb{Z}$-graded associative algebra

$$\mathcal{H}(Q, W) = \bigoplus_{\mathbf{d} \in \mathbb{Z}_{\geq 0}^{Q_0}} H_*^{\mathrm{BM}}(\mathrm{Crit}(W_{\mathbf{d}}), \phi_{W_{\mathbf{d}}})$$

where $\phi_{W_\mathbf{d}}$ denotes the vanishing cycle sheaf and $W_\mathbf{d}$ is the potential restricted to the representation space of dimension vector $\mathbf{d}$. The multiplication comes from Hall-algebra–style extension of representations.

15.3 $\mathbb{C}^3$ and the affine Yangian $Y(\widehat{\mathfrak{gl}}_1)$

Theorem 15.4 *(Schiffmann–Vasserot).* The critical CoHA of $(\mathbb{C}^3, 0)$ (Jordan quiver, cubic potential $W = \mathrm{tr}(XYZ - XZY)$) is isomorphic to the positive half of the affine Yangian of $\mathfrak{gl}_1$:

$$\mathcal{H}(\mathbb{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1).$$

The full affine Yangian $Y(\widehat{\mathfrak{gl}}_1)$ is isomorphic to the $\mathcal{W}_{1+\infty}$ algebra at the self-dual level — already a key object in the standard landscape of Volume I, where MC4 is solved by weight stabilization. The Yangian R-matrix is the DK-0 shadow.

15.4 General toric CY3 and affine super Yangians

THEOREM 15.5 *(Rapcak–Soibelman–Yang–Zhao)*. For a toric CY3 X without compact 4-cycles, the critical CoHA $\mathcal{H}(Q_X, W_X)$ is isomorphic to the positive half of the affine super Yangian $Y(\widehat{\mathfrak{g}}_{Q_X})$ associated to the toric quiver.

The toric diagram determines the quiver, hence the super Lie algebra $\mathfrak{g}_{Q_X}$, hence the affine super Yangian. The toric fan IS the root datum of the quantum vertex chiral group $G(X)$.

15.5 The CoHA as E_1-sector

The critical CoHA is an associative algebra — an E_1-algebra. In the language of this work:

- The CoHA = the positive half of $G(X)$ = the E_1-chiral sector (the ordered part).
- The full quantum vertex chiral group $G(X)$ is E_2 (braided).
- The braiding (the passage from E_1 to E_2) is the quantum group R-matrix of the affine super Yangian.
- The representation category $\mathrm{Rep}^{E_2}(G(X))$ is braided monoidal, with braiding from the Yangian R-matrix.

This is the "tree-level" CY3 combinatorics: the CoHA captures the genus-0 curve counting (DT invariants as intersection numbers on Hilbert schemes), and the E$_1 structureencodestheassociative(ordered)OPE.$

15.6 Root datum from toric geometry

The generalized root datum $\mathcal{R}(X_\Sigma)$ of a toric CY3:

(i) **Lattice**: $\Lambda(X) = K_0(\mathrm{Coh}(X))$ with the Euler form.

(ii) **Real roots**: vertices of the toric diagram (compact divisors).

(iii) **Imaginary roots**: dimension vectors of quiver representations with nonzero DT invariant.

(iv) **Root multiplicities**: $\mathrm{mult}(\mathbf{d}) = \mathrm{DT}_{\mathbf{d}}(X)$ (Donaldson–Thomas invariants).

(v) **Weyl group**: symmetries of the toric diagram.

Chapter 16

Fukaya Categories

16.1 The Fukaya category of a CY manifold

Example 16.1 *(Elliptic curve).* For an elliptic curve E_τ, $\mathrm{Fuk}(E_\tau)$ is CY of dimension 1. The Hochschild homology $\mathrm{HH}_\bullet(\mathrm{Fuk}(E_\tau))$ recovers the de Rham cohomology of E_τ, and the cyclic structure encodes the symplectic pairing. The associated chiral algebra is the Heisenberg vertex algebra H_k at level $k = \mathrm{vol}(E_\tau)$.

16.2 CY 2-folds: K3 and abelian surfaces

16.3 CY 3-folds

16.4 Wrapped Fukaya categories

16.5 The quantum chiral algebra of a Fukaya category

Chapter 17

Derived Categories of CY Manifolds

17.1 $D^b(\mathrm{Coh}(X))$ for CY manifolds

17.2 Homological mirror symmetry and the chiral algebra

The homological mirror symmetry conjecture (Kontsevich) identifies $\mathrm{Fuk}(X) \simeq D^b(\mathrm{Coh}(X^\vee))$. Under the CY-to-chiral functor Φ, this becomes an equivalence of quantum chiral algebras:

$$A_{\mathrm{Fuk}(X)} \simeq A_{D^b(\mathrm{Coh}(X^\vee))}.$$

17.3 Exceptional collections and chiral algebras

17.4 Stability conditions and the chiral moduli

Chapter 18

Matrix Factorizations

18.1 Matrix factorizations and Landau–Ginzburg models

18.2 The CY structure on $\mathrm{MF}(W)$

18.3 Quantum chiral algebras from singularities

18.4 ADE singularities and W-algebras

The connection between ADE singularities and W-algebras provides a key test case for the CY-to-chiral functor. For the A_{N-1} singularity $W = x^N$ in one variable, $\mathrm{MF}(x^N)$ is CY of dimension -1 (formal), so the relevant setting is the two-variable singularity $W = x^N + y^2$, giving $\mathrm{MF}(W)$ of CY dimension 0 (semisimple). To reach a CY_2 category related to W_N, one considers the threefold singularity $W = x^N + y^2 + z^2 + w^2$ in four variables, yielding $\mathrm{MF}(W)$ of CY dimension 2.

Chapter 19

Quantum Group Representations

19.1 $\mathrm{Rep}_q(\mathfrak{g})$ as a braided monoidal category

19.2 The quantum chiral algebra of $U_q(\mathfrak{g})$

Conjecture 19.1 *(Quantum group realization — Conjecture CY-C).* For a simple Lie algebra $\mathfrak{g}$ and $q = e^{\pi i/(k+h^\vee)}$ with k a positive integer, the braided monoidal category $\mathrm{Rep}_q(\mathfrak{g})$ should arise as $\mathrm{Rep}^{E_2}(A_C)$ for a CY category $C = C(\mathfrak{g}, q)$. The quantum chiral algebra A_C should recover the affine Kac–Moody vertex algebra $V_k(\mathfrak{g})$ at level k.

Status: The CY category $C(\mathfrak{g}, q)$ is not yet constructed in general. At generic q (irrational k), $\mathrm{Rep}_q(\mathfrak{g})$ is semisimple and the story simplifies; the interesting structure is at roots of unity, where the Kazhdan–Lusztig equivalence provides the prototype.

19.3 Kazhdan–Lusztig equivalence

The Kazhdan–Lusztig equivalence

$$\mathrm{Rep}_q(\mathfrak{g}) \simeq \mathrm{KL}_k(\mathfrak{g})$$

between quantum group representations (at root of unity) and the Kazhdan–Lusztig category of affine Kac–Moody modules is the prototype for Theorem CY-C.

19.4 Yangian and RTT realizations

Part VI

Connections and Frontier

Chapter 20

The Bar-Cobar Bridge to Volume I

20.1 CY bar complex vs chiral bar complex

The cyclic bar complex $\mathrm{CC}_\bullet(\mathcal{C})$ of a CY category and the chiral bar complex $B(A)$ of the associated chiral algebra are related by a canonical quasi-isomorphism (Theorem CY-A(ii)). This section spells out the dictionary.

20.2 Shadow tower of a CY category

20.3 Koszul duality: CY vs chiral

20.4 The five theorems in the CY setting

Chapter 21

Modular Koszul Duality and CY Geometry

21.1 The modular convolution algebra for CY categories

21.2 CY complementarity

21.3 The CY shadow CohFT

Chapter 22

Geometric Langlands and CY Quantum Groups

22.1 The geometric Langlands programme

22.2 CY categories in the Langlands correspondence

22.3 Quantum geometric Langlands and E_2-chiral algebras

22.4 The Feigin–Frenkel center and the Drinfeld center

Appendix A

Conventions

A.1 Grading conventions

All grading is **cohomological**: differentials have degree $+1$. The bar construction uses desuspension (matching Volumes I–II).

A.2 Sign conventions

Signs follow the Koszul rule throughout. For the curved A_∞-structure: $\mu_1^2(a) = [\mu_0, a]$ (commutator, minus sign).

A.3 Categorical conventions

- **dgCat** denotes the ∞-category of small dg categories over k.
- CY_d-**Cat** denotes the ∞-category of smooth d-dimensional CY categories.
- Tensor products of dg categories are derived unless stated otherwise.
- "Equivalence" means quasi-equivalence of dg categories (equivalence of associated ∞-categories).

A.4 Operadic conventions

- E_n denotes the little n-disks operad (Boardman–Vogt).
- $\overline{\mathrm{FM}}_k(\mathbb{C})$ denotes the Fulton–MacPherson compactification of $\mathrm{Conf}_k(\mathbb{C})$. Note: $\overline{\mathrm{FM}}$ is a blowup along diagonals, NOT the complement of the diagonal (AP, Vol I).
- $\mathrm{SC}^{\mathrm{ch,top}}$ denotes the Swiss-cheese operad from Volume II.

A.5 Bar complex and Koszul duality conventions

- $B(A)$ is the bar complex, a factorization *coalgebra* on $\mathrm{Ran}(X)$.

- $\Omega(B(A)) \simeq A$ is bar-cobar *inversion*: it recovers the original algebra (Vol I, Theorem B). This is NOT Koszul duality.
- $A^! = (H^*(B(A)))^\vee$ is the Koszul dual *algebra* (linear dual of bar cohomology).
- $D_{\mathrm{Ran}}(B(A)) \simeq B(A^!)$ is the Verdier dual (Vol I, Theorem A, Convention conv:bar-coalgebra-identity).
- The collision r-matrix $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_A)$ has pole orders *one less* than the OPE, because the bar construction extracts residues along $d\log(z_i - z_j)$, which absorbs one power of $(z - w)$ (Vol I, AP19).